

Introduction to Probability

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

Powerball: Is a jackpot 50–50?

Choose five distinct white numbers from 1–69 (order does not matter), and one red number from 1–26.



An example ticket

Match all five white numbers and the red number to win the jackpot.

Source: NJ Lottery — Powerball rules and odds

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The actual chance for one play: **1 in 292,201,338**.

Two event labels do not mean two equally likely outcomes.

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Lecture structure

1. Outcomes and events

One result $\longrightarrow$ all results $\longrightarrow$ a subset of results.

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3. Event operations

NOT, OR, AND $\rightarrow$ complement, union, intersection.

SECTION 1

Probability models

Sample space and event

MUST UNDERSTAND

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Individual outcome ω One result: a ticket number, e.g. $\omega = 42$.

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The event occurs exactly when $\omega \in E$.

$$7 \in E \quad (\text{odd}), \quad 42 \notin E \quad (\text{even}).$$

Classical probability model

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Let S be **finite and nonempty**. The classical model **assumes every outcome in S is equally likely**.

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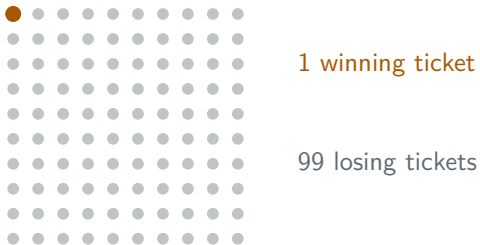
$$0 \leq \Pr[E] \leq 1, \quad \Pr[\emptyset] = 0, \quad \Pr[S] = 1.$$

Classical probability: Rule of thumb

$$\Pr[\text{some event}] = \frac{\# \text{ outcomes in which the event happens}}{\# \text{ total different outcomes}}.$$

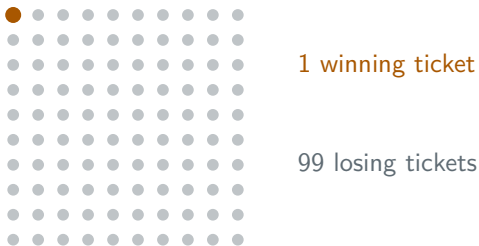
A simplified lottery: Count the outcomes

Only ticket 1 wins. Each of the 100 tickets is **equally likely**.



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$$\Pr[\text{jackpot}] = \frac{1}{100}, \quad \Pr[\text{no jackpot}] = \frac{99}{100}.$$

General discrete probability model

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For any event $E \subseteq S$,

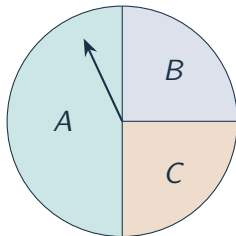
$$\Pr[E] = \sum_{x \in E} \Pr[x].$$

Add the probabilities of all outcomes in E .

Example: An unequal spinner

The spinner lands uniformly by angle. Record its label: $S = \{A, B, C\}$.

$$\Pr[A] = \frac{1}{2}, \quad \Pr[B] = \Pr[C] = \frac{1}{4}.$$

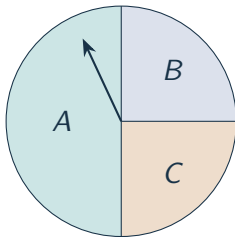


For $E = \{A, B\}$, what is $\Pr[E]$?

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For $E = \{A, B\}$, what is $\Pr[E]$?

$$\Pr[E] = \Pr[A] + \Pr[B]$$

$$= \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \neq \frac{2}{3}.$$

Roadmap: Three probability models

Return to **finite, equally likely outcomes**.

Model	Experiment and one outcome	Total outcomes
1. Independent trials	Flip a fair coin t times: an ordered string of results.	2^t

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2. Random permutation	Shuffle 52 distinct cards: a complete order of the deck.	$52!$
3. Drawing without replacement	Draw five cards from 52: a hand, ignoring order.	$\binom{52}{5}$

SECTION 2

Independent random trials

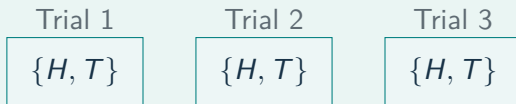
Model 1: Independent random trials

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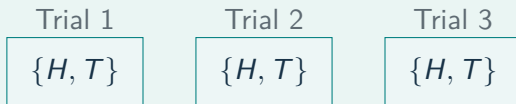


One outcome: *HTH*. Repeated results are allowed.

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$$|S| = m^t \quad (t \text{ independent trials, } m \text{ equally likely choices each}).$$

Independent, uniform choices make all m^t sequences equally likely.

Example: Three coin flips

Flip a **fair coin** three times **independently**.

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Coin flips: Count the sequences

Three fair, independent flips: $2^3 = 8$ **equally likely outcomes**.

<i>HHH</i>	<i>HHT</i>	<i>HTH</i>	<i>HTT</i>
<i>THH</i>	<i>THT</i>	<i>TTH</i>	<i>TTT</i>

Event: $E = \{HHT, HTH, THH\}$. Equivalently, choose the two H positions.

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Event: $E = \{HHT, HTH, THH\}$. Equivalently, choose the two H positions.

$$|E| = \binom{3}{2} = 3, \quad \Pr[E] = \frac{\binom{3}{2}}{2^3} = \frac{3}{8}.$$

Unequal reduced outcomes

Two fair coin flips; the first result does not affect the second.

All four strings HH , HT , TH , TT are equally likely.

Keep only the number of heads. Are its three values equally likely?

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Head count	0	1	2
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Number of full outcomes	1	2	1
Probability	$1/4$	$1/2$	$1/4$

Example: Drawing with replacement

Draw from ten balls labelled $1, \dots, 10$, uniformly and independently.

Return the ball and mix after each draw.



What is the probability of drawing 3, 7, 1 in that order?

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One outcome is an ordered triple; repeated labels are allowed.

$$|S| = 10^3, \quad E = \{(3, 7, 1)\}, \quad \Pr[E] = \frac{1}{1000}.$$

SECTION 3

Random permutation

Model 2: Random permutation

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A **uniform random permutation** arranges every distinct object exactly once, with every complete order equally likely.

$$|S| = 52 \cdot 51 \cdots 1 = \mathbf{52!}.$$

In general: N distinct objects give $N!$ equally likely orders.

Example: One specific shuffle

Shuffle a standard **52-card deck, without jokers**, uniformly.

Specify one **complete card order** ω_0 before shuffling.

$$\omega_0 = (A\heartsuit, 2\clubsuit, 7\spadesuit, \dots) \quad (\text{beginning shown})$$

What is the probability of exactly this order?

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What is the probability of exactly this order?

$$\Pr[\omega_0] = \frac{1}{52!}.$$

Example: The first ace

Shuffle a standard 52-card deck. All $52!$ orders are equally likely.

Reveal cards one by one until an ace appears.



What is the probability that the first ace is $A\heartsuit$?

Solution 1: Direct counting

$$\Pr[\text{first ace is } A\heartsuit] = \frac{\# \text{ orders whose first ace is } A\heartsuit}{\# \text{ total different orders}}.$$

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$$\begin{aligned}\Pr[\text{first ace is } A\heartsuit] &= \frac{\# \text{ orders whose first ace is } A\heartsuit}{\# \text{ total different orders}}. \\ &= \frac{\# \text{ orders whose first ace is } A\heartsuit}{52!}.\end{aligned}$$

We only need to count the numerator.

Direct counting: Count the numerator



Choose the four ace positions.

$$\binom{52}{4}$$

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Choose the four ace positions. $\binom{52}{4}$

Put $A\heartsuit$ in the earliest; permute the other three aces. $3!$

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Put $A\heartsuit$ in the earliest; permute the other three aces. $3!$

Arrange the 48 non-aces in the remaining positions. $48!$

$$|E| = \binom{52}{4} 3! 48!,$$

Direct counting: Count the numerator



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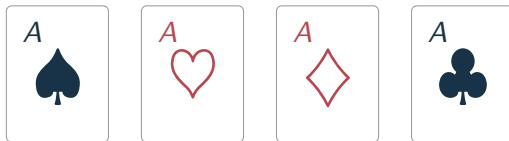
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$$|E| = \binom{52}{4} 3! 48!, \quad \Pr[E] = \frac{\binom{52}{4} 3! 48!}{52!} = \frac{1}{4}.$$

Solution 2: Relative counting

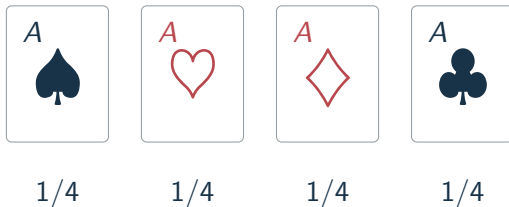
Ignore the non-aces. Which of the four aces comes first?



In a uniform shuffle, no ace is favored: all four are equally likely to come first.

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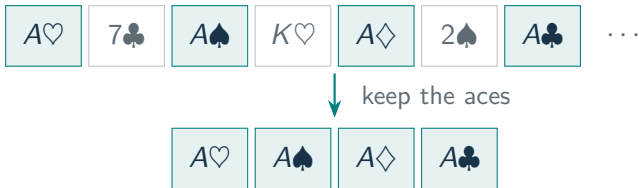
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$$\Pr[\text{first ace is } A\heartsuit] = \frac{1}{4}.$$

Why relative counting works

Start with a **uniform shuffle**. Fix any subset of k cards.

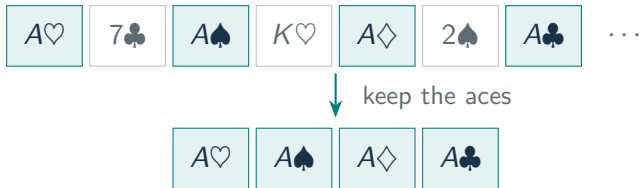
Keep only those cards: their $k!$ **relative orders are equally likely**.



Why relative counting works

Start with a **uniform shuffle**. Fix any subset of k cards.

Keep only those cards: their $k!$ **relative orders are equally likely**.



Swapping the selected cards' labels pairs the possible orders; a uniform shuffle gives each the same probability.

For the four aces: $4!$ equally likely orders, and each ace is first in $3!$.

Example: The first ace or king

Shuffle the 52-card deck uniformly. Reveal cards one by one.

What is the probability that the first ace appears before the first king?



First ace before first king



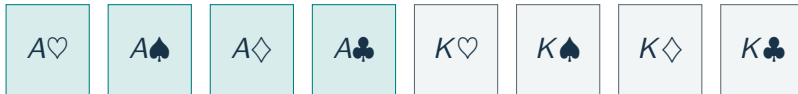
First king before first ace

Only aces and kings are shown.

Aces and kings: Which comes first?

Keep only the **four aces and four kings**. Their relative order is uniform.

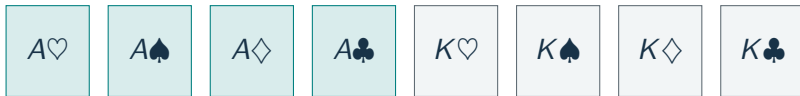
Each of these eight cards is equally likely to come first.



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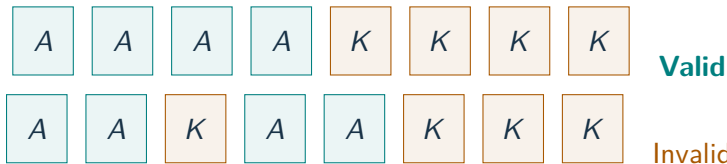


$$\Pr[\text{first ace before first king}] = \Pr[\text{first of these eight is an ace}] = \frac{4}{8} = \frac{1}{2}.$$

Example: All aces before the first king

Shuffle the 52-card deck uniformly. Reveal cards one by one.

What is the probability that all four aces appear before the first king?



Only aces and kings are shown; other cards may occur between them.

All aces: Count eight-card orders

Keep only the **four distinct aces and four distinct kings**. Their $8!$ **relative orders are equally likely**.

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Order the four aces: $4!$ Order the four kings: $4!$

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$$\Pr[\text{all aces before the first king}] = \frac{4! 4!}{8!} = \frac{1}{\binom{8}{4}} = \frac{1}{70}.$$

SECTION 4

Random drawing without replacement

Model 3: Drawing without replacement

Select r of N distinct objects uniformly, **without returning them**.

Record **which objects**, not their draw order.

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Example: select three labels from $\{A, B, C, D, E\}$.



One outcome: $\{A, C, E\}$

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Example: select three labels from $\{A, B, C, D, E\}$.



$$|S| = \binom{N}{r} \quad (1 \leq r \leq N).$$

Each subset has $r!$ equally likely draw orders, so the subsets are equally likely.

Example: One specific hand

Draw five cards uniformly **without replacement**. Ignore their order.

What is the probability of this particular hand H_0 ?



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All $\binom{52}{5}$ hands are equally likely: each has $5!$ draw orders.

$$\Pr[\{H_0\}] = \frac{1}{\binom{52}{5}} = \frac{1}{2,598,960}.$$

Example: Four of a kind

Draw five cards uniformly **without replacement**. Ignore their order.

What is the probability of four of a kind?

Four cards of the same rank, plus one other card. For example:



Four of a kind: Count the hands

All equally likely hands: $|S| = \binom{52}{5}$.

Construct a qualifying hand	Choices
Choose the four-card rank: $2, 3, \dots, 10, J, Q, K, A$.	13
Take all four cards of that rank.	1

Four of a kind: Count the hands

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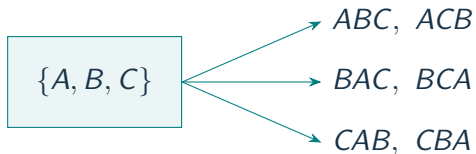
$$\Pr[\text{four of a kind}] = \frac{13 \cdot 1 \cdot 48}{\binom{52}{5}} = \frac{624}{2,598,960} \approx \mathbf{0.0002401}.$$

Without replacement: When order matters

A hand ignores order. A question about **first or second** must keep it.

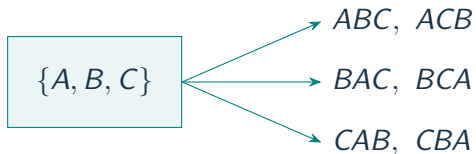
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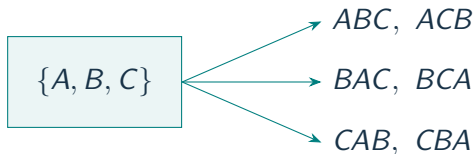


Choose the subset, then its order:

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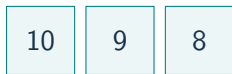
$$|S_{\text{ordered}}| = \binom{N}{r} r! = \frac{N!}{(N-r)!}.$$

Equivalently, reveal the first r objects of a uniform random permutation. Each ordered sample has $(N-r)!$ completions.

Example: Drawing without replacement

Draw three balls from labels $1, \dots, 10$, uniformly **without replacement**.

What is the probability of drawing 3, 7, 1 in that order?

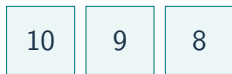


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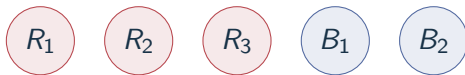


Choices at draws 1, 2, 3.

$$|S| = 10 \cdot 9 \cdot 8 = 720, \quad |E| = 1, \quad \Pr[E] = \frac{1}{720}.$$

Practice: Red then blue

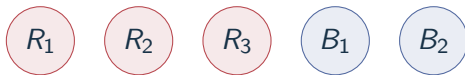
A bag contains three red and two blue balls. Draw two uniformly **without replacement**.



What is the probability of red first, then blue?

Practice: Red then blue

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What is the probability of red first, then blue?

$$|S| = 5 \cdot 4, \quad |E| = 3 \cdot 2, \quad \Pr[E] = \frac{3 \cdot 2}{5 \cdot 4} = \frac{3}{10}.$$

Three models: Choose the sample space

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Uniform drawing without replacement	An r -element subset; order ignored	$\binom{N}{r}$

If draw order matters in the last model: $\binom{N}{r} r! = N! / (N - r)!$.

SECTION 5

Event operations

Event operations and set operations

MUST UNDERSTAND

Let $A, B \subseteq S$ be events in the same sample space.

An event is a set of outcomes, so we combine events using set operations.

Event operation	Corresponding set	When it occurs
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NOT A	Complement: $\bar{A} = S \setminus A$	A does not occur.
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A OR B	Union: $A \cup B$	At least one occurs (possibly both).
A AND B	Intersection: $A \cap B$	Both occur.

One card, two events

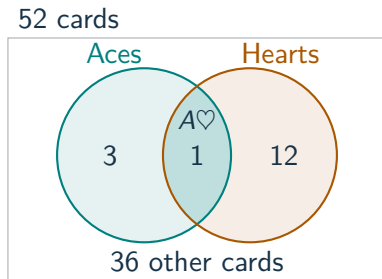
Draw one card uniformly from a standard 52-card deck.

A : an ace (4 cards), B : a heart (13 cards).

One card, two events

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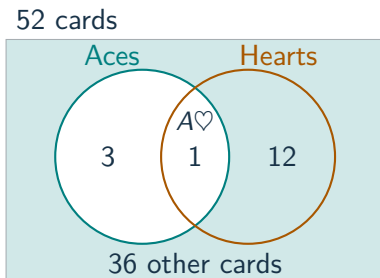
A : an ace (4 cards), B : a heart (13 cards).



NOT: Not an ace

Draw one card uniformly from a standard 52-card deck.

What is the probability that it is NOT an ace?



A : an ace. $\bar{A}$: not an ace.

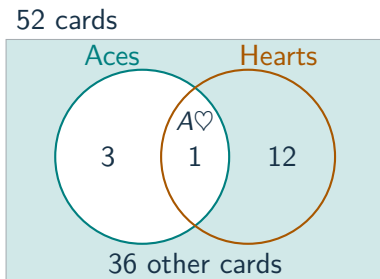
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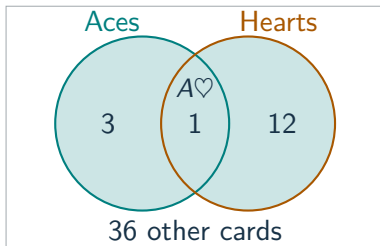
$$\Pr[\text{not an ace}] = 1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}.$$

OR: An ace or a heart

Draw one card uniformly from a standard 52-card deck.

What is the probability that it is an ace OR a heart?

52 cards



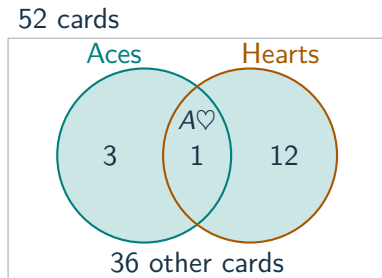
A : an ace. B : a heart.

Which card is counted twice in $4 + 13$?

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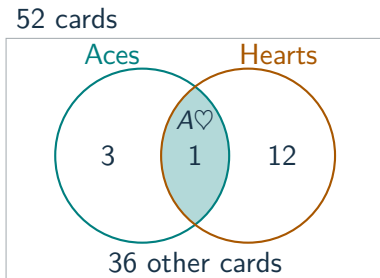
$$A \cap B = \{A\heartsuit\}.$$

$$\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B] = \frac{4 + 13 - 1}{52} = \frac{4}{13}.$$

AND: An ace and a heart

Draw one card uniformly from a standard 52-card deck.

What is the probability that it is an ace AND a heart?



A : an ace.

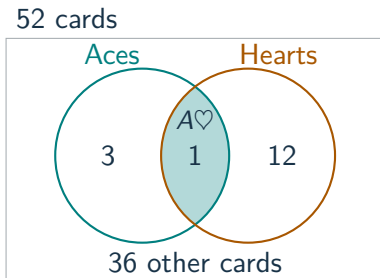
B : a heart.

Which card satisfies both conditions?

AND: An ace and a heart

Draw one card uniformly from a standard 52-card deck.

What is the probability that it is an ace AND a heart?



A : an ace. B : a heart.

Which card satisfies both conditions?

Only the ace of hearts:

$$A \cap B = \{A♥\}, \quad \Pr[A \cap B] = \frac{1}{52}.$$

AND means intersection: count the outcomes satisfying both.

Five-card hand: At least one ace

Draw five cards uniformly **without replacement**; ignore order.

What is the probability of at least one ace?



One outcome is now a **hand**: $|S| = \binom{52}{5}$.

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Complement: no aces.

At least one ace: Count the complement

A hand with no aces uses only the **48 non-aces**.



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A hand with no aces uses only the **48 non-aces**.



$$\Pr[\text{no aces}] = \frac{\binom{48}{5}}{\binom{52}{5}}.$$

At least one ace: Count the complement

A hand with no aces uses only the **48 non-aces**.



$$\Pr[\text{no aces}] = \frac{\binom{48}{5}}{\binom{52}{5}}.$$

$$\Pr[\text{at least one ace}] = \mathbf{1} - \frac{\binom{48}{5}}{\binom{52}{5}}.$$

Example: An ace and a heart

Draw five cards uniformly **without replacement**; ignore order.

What is the probability of at least one ace AND at least one heart?



The $A\heartsuit$ satisfies both requirements.

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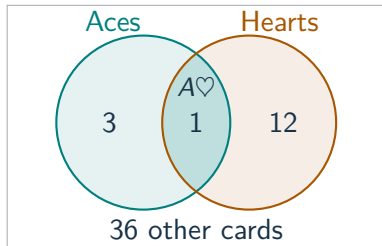
Which hands fail?

No aces **OR** no hearts.

Example: Count the bad hands

N_A : no aces. N_H : no hearts.

52 cards



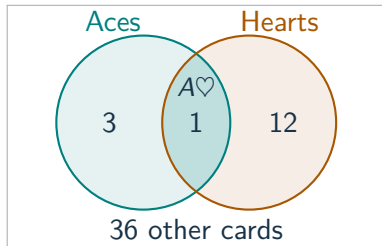
Counts of individual cards

Bad hands	Number
N_A : choose from 48 non-aces	$\binom{48}{5}$

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Counts of individual cards

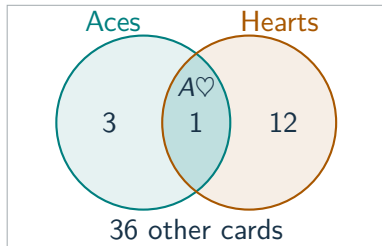
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N_A : choose from 48 non-aces	$\binom{48}{5}$
N_H : choose from 39 non-hearts	$\binom{39}{5}$

Example: Count the bad hands

N_A : no aces.

N_H : no hearts.

52 cards



Counts of individual cards

Neither an ace nor a heart: $52 - 4 - 13 + 1 = 36$.

Bad hands

Number

N_A : choose from 48 non-aces $\binom{48}{5}$

N_H : choose from 39 non-hearts $\binom{39}{5}$

$N_A \cap N_H$: choose from 36 cards $\binom{36}{5}$

Example: Complement and inclusion–exclusion

E : at least one ace and at least one heart.

$$E = S \setminus (N_A \cup N_H).$$

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$$\Pr[E] = 1 - \Pr[N_A] - \Pr[N_H] + \Pr[N_A \cap N_H].$$

$$\Pr[E] = 1 - \frac{\binom{48}{5} + \binom{39}{5} - \binom{36}{5}}{\binom{52}{5}}.$$

Practice

For each question: define S , identify E , then calculate $\Pr[E]$.

1. Two fair dice with 36 equally likely pairs: a sum of 9?
2. A uniformly selected five-bit string: at least one 1?
3. One uniformly selected card: a spade or an ace?

Practice

Event	Count or correction	Probability
Sum 9	4 pairs out of 36	$1/9$
At least one 1	Exclude 00000 from 32 strings	$31/32$
Spade or ace	$13 + 4 - 1$ cards out of 52	$4/13$

Challenge exercise: Nondecreasing ranks

Draw five cards uniformly **without replacement**, keeping their order.

What is the probability that their ranks are nondecreasing?

$$A < 2 < 3 < \dots < 10 < J < Q < K.$$

Equal ranks are allowed; suits do not affect the rank comparison.

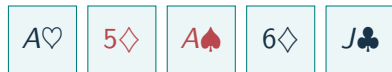
Valid



$$A \leq A \leq 5 \leq 6 \leq J.$$

Solution in the course note.

Invalid



$5 > A$: the rank decreases.

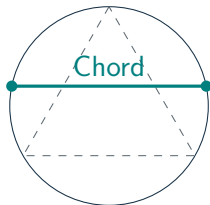
Probability summary

Concept	Definition or rule
Sample space and event	S : all outcomes; $E \subseteq S$: desired outcomes.
Classical probability	$\Pr[E] = E / S $, for finite nonempty S with equally likely outcomes.
General discrete model	$\Pr[E] = \sum_{x \in E} \Pr[x]$.
Reduce outcomes	Keep what determines E ; equal full-outcome counts preserve uniformity.
Event operations	NOT: $\Pr[\bar{E}] = 1 - \Pr[E]$. OR: $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$. AND: the event $A \cap B$.

Optional: Bertrand's paradox

Choose a **random chord** of a circle.

Is it longer than a side of the inscribed equilateral triangle?



Dashed: equilateral triangle

Infinitely many chords: no finite total to count.

What does “random” mean?

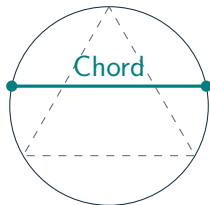
Two endpoints chosen independently, uniformly on the circle

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Two endpoints chosen independently, uniformly on the circle $1/3$

Midpoint chosen uniformly by area inside the disk $1/4$